

Applied AI Scientist | Generative & Multimodal AI · Vision-Language Models · LLM Applications
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SARAWUT OUNJAI, PhD

DPST Scholar, Development and Promotion of Science and Technology Talents Project, Royal Thai Government

PROFILE

Applied AI scientist with a PhD from Imperial College London, building vision-language, multimodal retrieval and LLM-based systems that solve real operational problems. I take AI work from problem framing through experimentation to a validated, documented system, and I set the evaluation standard that decides whether a result is real: held-out benchmarks, bootstrap confidence intervals, ablation studies and error analysis.

At Gosoft (CP ALL Group), I set the technical direction for AI initiatives serving one of Southeast Asia's largest retail networks, spanning shelf-compliance vision, cross-retailer price intelligence with multimodal LLMs, LLM-based quality assurance, CCTV analytics and Thai-language document retrieval. I package this work so other teams can adopt it: four reusable AI capabilities, with architecture documents, cost models, statements of work and user manuals, now serve as building blocks across the organisation. I translate cost and accuracy trade-offs for executives, and coach engineering students and interns in rigorous, reproducible practice.

Signature Results

- Cross-retailer promotion matching with multimodal LLM reranking: **95.7% end-to-end top-1 accuracy with zero false rejections**; manuscript being finalised for Scientific Reports.
- Shelf-compliance vision at **97.3% detection recall** and **94.4% top-1 SKU identification**.
- Adapted a 2B-parameter vision-language retriever to Thai documents with LoRA, fully offline: **nDCG@10 0.482 → 0.923** on a single 8 GB GPU.
- Built a **local-first, 9-agent AI system** for data science work, with review loops and human approval gates.
- **23 planned deliverables closed on schedule** in H1/2026, each with a written evaluation report.

TECHNICAL SKILLS

- **Generative & Agentic AI:** multimodal LLM pipelines · multi-agent orchestration with review loops and approval gates · local LLM serving (Ollama) · hybrid LLM plus deterministic verification · prompt engineering · abstention design · token-cost modelling
- **Vision-Language & Retrieval:** SigLIP · DINOv3 · ColQwen2 / ColPali · FAISS · BM25 and hybrid lexical-dense search · LightGBM reranking · contrastive and triplet learning · hard-negative mining
- **Deep Learning & Vision:** PyTorch · TensorFlow · Hugging Face · LoRA / PEFT fine-tuning · GPU training and inference · YOLO11 · RT-DETR / RF-DETR · SAM · SAHI · OpenCV
- **Evaluation & Statistics:** nDCG / Recall@k / mAP · paired bootstrap confidence intervals · ablation and error analysis · ground-truth and experimental design · XGBoost
- **Engineering & MLOps:** Python · R · SQL · Docker · Kubernetes · Argo Workflows · MLflow · Bitbucket Pipelines · Git · Streamlit · AWS · HPC · ETL pipelines
- **Enablement & Communication:** reusable components and reference pipelines · architecture and SOW authoring · user manuals and knowledge-transfer material · executive briefings · technical coaching · scientific writing

PROFESSIONAL EXPERIENCE

Data Scientist, Gosoft (Thailand) Co., Ltd. (CP ALL Group) Nov 2025 – present
Data Science & Data Engineering Unit, IT Service · Chiang Mai, Thailand

- **Technical direction:** set the approach for the unit's AI initiatives across five systems in retail vision, retrieval and LLM applications, covering model and architecture selection, experimental design, and the criteria a system must meet before it is judged ready.
- **Internal enablement:** delivered four reusable AI capabilities adopted by other teams (store-image detection, SKU image retrieval, hybrid LLM verification, cross-store product matching), cutting duplicated engineering effort on each new use case; authored the architecture documents, cloud and LLM cost models, statements of work and user manuals that made adoption possible.

- **Evaluation standards:** established the unit's benchmarking practice (ground-truth construction, held-out benchmarks, paired bootstrap confidence intervals, ablation and error analysis), so decisions rest on measured evidence rather than single favourable runs.
- **Modelling:** fine-tune vision and vision-language models (SigLIP, DINOv3, ColQwen2) with LoRA and contrastive objectives; build retrieval stacks combining FAISS, BM25 and learned rerankers; apply multimodal LLMs to rerank image-and-text candidates; design hybrid architectures pairing LLM reasoning with deterministic checks for auditable decisions.
- **Infrastructure:** reproducible GPU training and batch inference on Docker, Argo Workflows over Kubernetes, Bitbucket pipelines, MLflow tracking, and AWS/HPC compute.
- **Stakeholders:** translate business requirements into solutions with cross-functional teams and present cost–accuracy trade-offs to non-technical and executive audiences.
- **Delivery record:** closed 23 planned deliverables on schedule in H1/2026, each with a written sprint or evaluation report (58 technical documents).

FLAGSHIP AI PROJECTS

Production: Cross-Retailer Promotion Price Comparison: Vision + NLP + Multimodal LLM (2026)

- Designed a five-stage retrieve-then-rerank pipeline (acquisition → promotion detection → text extraction → embedding retrieval → multimodal LLM reranking) matching competitor promotions to equivalent in-house products.
- Retriever reached Acc@1 80.7% and Recall@10 97.2%, with a soft price signal adding 6.8 points at no LLM cost; multimodal LLM reranking lifted end-to-end Acc@1 to 95.7%.
- Engineered an abstention mechanism at a 98.0% true-negative rate with zero false rejections, so unmatched promotions are flagged rather than mismatched.
- Improved the vision backbone through staged experiments: LoRA fine-tuned SigLIP/DINOv3, projection-head adaptation, hard-negative mining, and a LightGBM reranker under GroupKFold protocols. Manuscript being finalised for submission to Scientific Reports.

Production: Product-on-Shelf Compliance: Computer Vision (2026)

- Built a system that detects products in a shelf photograph, identifies each SKU and checks the arrangement against the planogram, delivered as a two-mode application for catalogue registration and compliance checking.
- 97.3% filtered detection recall with YOLO11, benchmarked against RT-DETR/RF-DETR with validation-versus-test gap analysis; 94.4% top-1 and 98.6% top-3 SKU identification with a LoRA fine-tuned SigLIP encoder and FAISS.
- Needleman–Wunsch alignment for shelf ordering with DBSCAN row grouping and perspective correction; 79.7% end-to-end accuracy on genuine in-store imagery, with failure modes attributed by stage.

Production: CCTV Revenue Verification (2026)

- RF-DETR with frame sampling and temporal voting: 100% video-level detection accuracy across 15 held-out videos (frame mAP50 = 0.96), no cases requiring manual review; supervised an intern through the full experimental cycle.

PoC: LLM Quality Assurance for Marketing Artwork (2026)

- Paired an LLM with deterministic numeric checks for auditable pass/fail decisions; built the token-cost model and authored the SOW and design document.

PoC: Thai-Language Document Retrieval for Government Forms (2026)

- Compared OCR-based and OCR-free retrieval for Thai government forms on a benchmark of 84 real documents (7 types, 111 queries), with 10,000-iteration paired bootstrap confidence intervals on every comparison.
- Adapted a 2B-parameter late-interaction vision-language retriever (ColQwen2) to Thai with LoRA, trained on 4,910 fully synthetic documents and 34,417 generated queries on one 8 GB laptop GPU in 5.3 hours, keeping all data offline: nDCG@10 rose from 0.482 to 0.923 (+44.1 points, 95% CI +35.4 to +52.8).
- Validated on unseen document types, five query paraphrase styles and genuine photographed documents. Key finding: rebalancing the training query mix added +35.3 points at zero cost, while a seven-fold data increase added a non-significant +1.8.
- Recommended an OCR-plus-lexical route for deployment (nDCG@10 = 0.981, 3.1 KB/page index, no GPU at serving time), delivered as a technical report and executive brief with cost models.

INDEPENDENT PROJECTS

Alpha-AI: Local-First Multi-Agent Data Science System

- Built a local-first multi-agent system that runs end-to-end data science workflows, covering data intake, EDA, feature engineering, modelling, evaluation, reporting and academic paper writing. Nine specialised agents coordinate through an automated review loop and human approval gates, powered by local LLMs via Ollama so that data never leaves the machine.

LEADERSHIP & MENTORING

- **Technical coach, Gosoft × Chulalongkorn University Capstone Programme (2026):** guided two teams of computer engineering students (10 in total) on "AI Skin Analyser" and "AI Product on Shelves" through problem formulation, dataset and ground-truth design, methodology and evaluation; reviewed final reports with point-by-point written feedback.
- **Mentor, data science internship programme (2026):** supervised an intern through a full computer-vision proof of concept, from experimental design and annotation to evaluation and reporting; coached interns from several universities on reproducible working practice.
- **Knowledge transfer:** produced architecture documentation, pipeline knowledge-transfer material with diagrams and a user manual, contributing to seven items on the organisational IT roadmap in H1/2026, so later projects start from proven components.

RESEARCH EXPERIENCE

Postdoctoral Researcher (part-time), Naresuan University & Natural History Museum
May 2024 – present

- Developing a beetle classification model for the Natural History Museum's SITE-100 programme, alongside the Gosoft role; manuscript in preparation.
- First-author manuscript on a DNA-based inventory of Thai Coleoptera submitted to Insect Systematics and Diversity.

Postdoctoral Researcher, Chiang Mai University, Thailand 2023–2024

- Environmental-DNA assays for invasive fish in Thai rivers supporting national invasive-species management (*iScience*, *NeoBiota*, 2026).

Doctoral Research, Imperial College London & Natural History Museum, UK 2018–2023

- Built automated HPC pipelines for sequence retrieval, BLAST, alignment, tree construction and data management, processing over 170,000 biological samples; developed machine-learning methods for sequence authentication and image-based specimen classification.
- Integrated 10,466 sequences into a global phylogenetic analysis of tropical forest beetle communities; first author on a phylogenetic authentication method in *Molecular Ecology Resources*.

Earlier Research, Chiang Mai University, Thailand 2014–2017

- DNA barcoding with high-resolution melting for authentication of Thai herbal products (*Proc. R. Soc. B*).

EDUCATION

PhD in Life Sciences | Imperial College London, United Kingdom | 2018–2023
M.Sc. in Biology | Chiang Mai University, Thailand | 2013–2015 | GPA 4.00
B.Sc. in Biology | Chiang Mai University, Thailand | 2009–2012 | GPA 3.60, First-Class Honours
Awards: Academic Excellence Award · DPST scholarship (Royal Thai Government)

PUBLICATIONS & IMPACT

11 peer-reviewed publications (190 citations, h-index 4); full list on Google Scholar.

Selected

- Ounjai, S., et al. (2026). Phylogenetic authentication of amplicon sequence variants in single-specimen metabarcoding of tropical insects. *Molecular Ecology Resources*.
- Creedy, T. J., … Ounjai, S., et al. (2026). 2,366 new mitochondrial genomes with preliminary identification and phylogeny of >5,500 putative species of beetles. *Scientific Data*.
- de Boer, H. J., … Ounjai, S., et al. (2017). DNA metabarcoding of orchid-derived products reveals widespread illegal orchid trade. *Proceedings of the Royal Society B*, 284(1863).

Submitted

- Ounjai, S., et al. (submitted). A DNA-based inventory of Coleoptera in Thailand. *Insect Systematics and Diversity*.

TRAINING & COURSES

- **AI:** AI for All (NVIDIA) · Claude 101 (Anthropic) · Generative AI for Products and Creative Work (Big Data Institute) · Elements of AI (University of Helsinki) · 35 courses and 80+ learning hours on the corporate platform.
- **AI governance (public sector):** AI Governance & Ethics and Prompt Engineering (Digital Government Academy) · AI Ethics and Research Integrity (NRCT).
- **International research training:** Uppsala University, Sweden (2015, 5 months) · CERTH, Thessaloniki, Greece (2012).
- References available on request.